



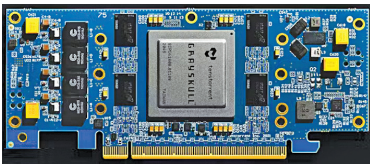
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

ML-acceleration Grayskull development board

Tenstorrent has launched its first Grayskull development board, which the company claims to offer high-efficiency machine learning (ML) acceleration.



The board's processor is made up of a grid of Tensix cores, each of which is built from five process cores around the RISC-V instruction set architecture. The Grayskull e150 offers performance from 120 Tensix cores running at a higher 1.2GHz clock speed. These additional cores mean a total of 120MB of SRAM, though the board only offers the same 8GB of LPDDR4. They also result in a higher power draw, with the card pulling 200W at peak load. The board is designed for those who want to 'run models right away,' offering a top-down approach that takes existing models from frameworks like PyTorch and TensorFlow and makes them executable on the Grayskull hardware.

Tenstorrent
<https://tenstorrent.com>

MPU for vision AI and real-time control

Renesas has launched the RZ/V2H, which the company claims to offer high performance within the family and enables vision AI and real-time control capabilities. The DRP-AI3 accelerator's pruning technology enhances AI computing efficiency, increasing AI inference performance up to 80 TOPS. This allows engineers to process vision

AI applications at the edge without relying on cloud computing. The processor includes four Arm Cortex-A55



CPU cores at 1.8GHz for Linux applications, two Cortex-R8 cores at 800MHz for real-time processing, and one Cortex-M33 sub-core. This combination allows the MPU to manage vision AI and real-time control tasks, suitable for robotics applications. Its power consumption reduces the need for cooling fans and components, enabling smaller, cost-effective, and reliable system designs.

Renesas
<https://www.renesas.com>

3D lidar module for on-chip dToF processing

STMicroelectronics's VL53L9 is a direct time-of-flight (dToF) 3D lidar module with a 2.3k resolution, featuring dual-scan flood illumination for object detection and capturing 2D infrared images and 3D depth maps. It is a low-power



module that comes ready to use with on-chip dToF processing and requires no additional external components or calibration.

The device also offers ranging performance from 5cm to 10 metres. It enables features such as laser autofocus, bokeh, and cinematic effects for both still and video at 60fps. Virtual reality systems can use the sensor's depth and 2D images for improved spatial mapping, leading to immersive gaming and VR experiences.

STMicroelectronics
<https://www.st.com>

Dual varistor for ADAS in vehicles

TDK Corporation has launched the AVRH16A2C270KT200NA8 for automotive applications. The varistor meets the electrostatic discharge (ESD) requirements for the operation of advanced driver assistance systems (ADAS) in



vehicles. It supports electronics on a CAN bus and has a 2-in-1 array structure, integrating the functions of two varistors into

a single element. It is designed at the 1608 size (1.6mm x 0.8mm x 0.6mm) and minimises the capacitance difference between channels with TDK's design technology. The main features of the AVRH series varistors include ESD resistance of 25kV, an operating temperature range of up to 150°C, and compliance with the AEC-Q200 standard. The model has a 2-in-1 array structure that provides dual varistor functionality and minimises the capacitance difference between channels to less than 1.0pF.

TDK
<https://www.tdk.com>

Industry-first 12-stack HBM3E DRAM

Samsung Electronics has announced the development of HBM3E 12H, the first 12-stack HBM3E DRAM and the



highest-capacity HBM product so far. The HBM3E 12H offers bandwidth of up to 1280GB/s and a capacity of 36GB, both more than 50% higher than the 8-stack HBM3 8H. This technology is expected